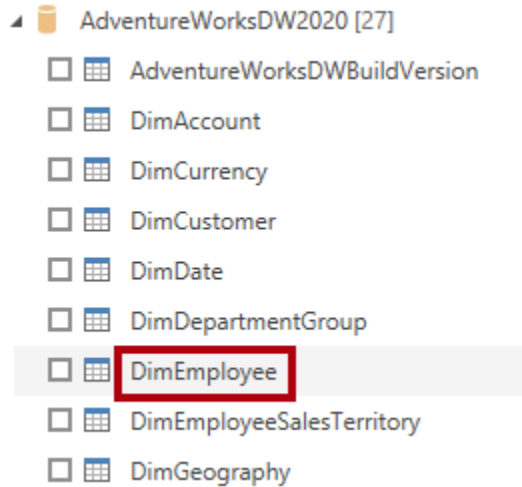
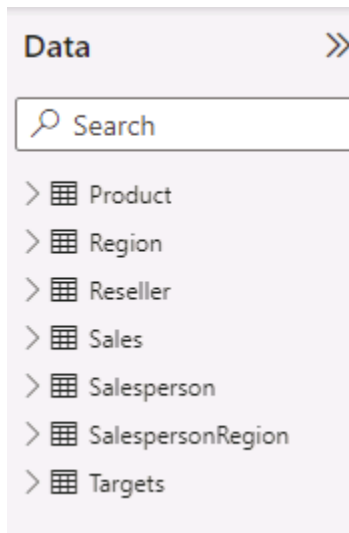


[PL-300-Microsoft-Power-BI-Data-Analyst/Instructions/Labs/01-get-data-in-power-bi.md at Main · MicrosoftLearning/PL-300-Microsoft-Power-BI-Data-Analyst](#)

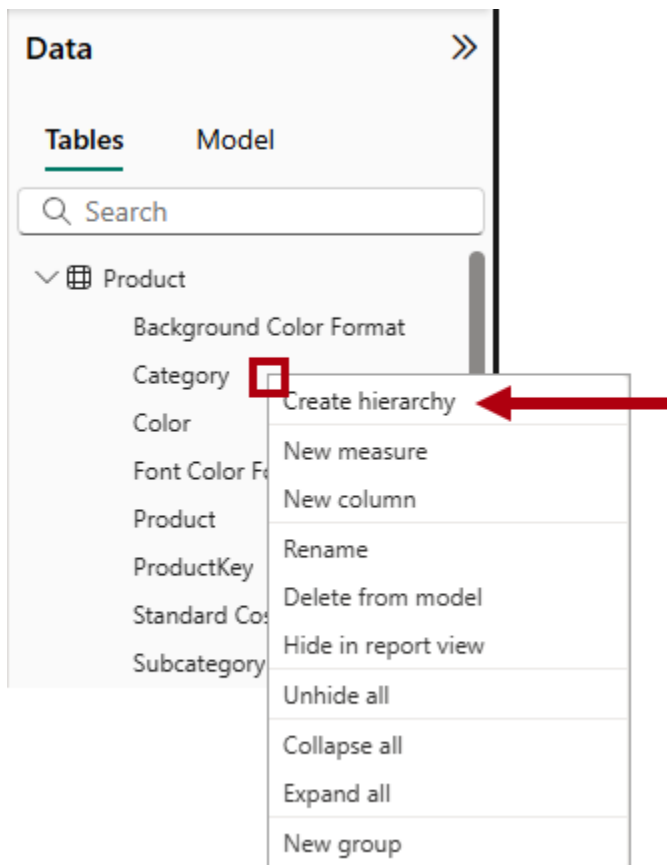
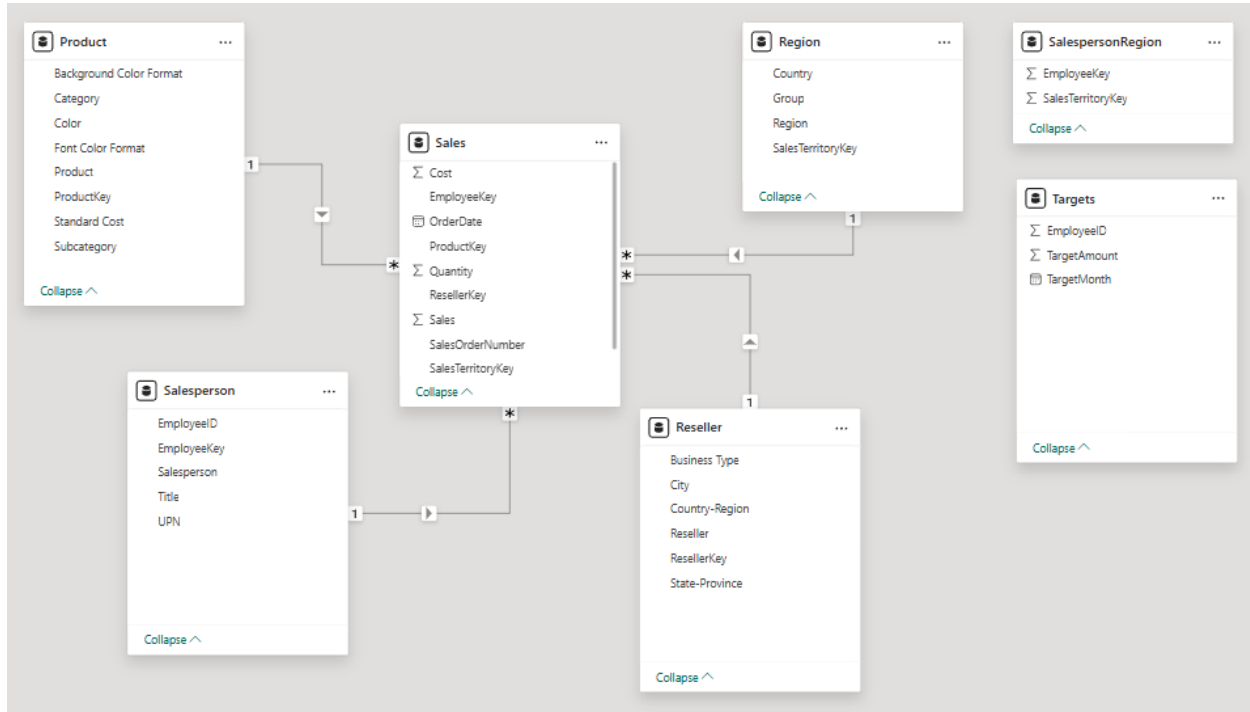


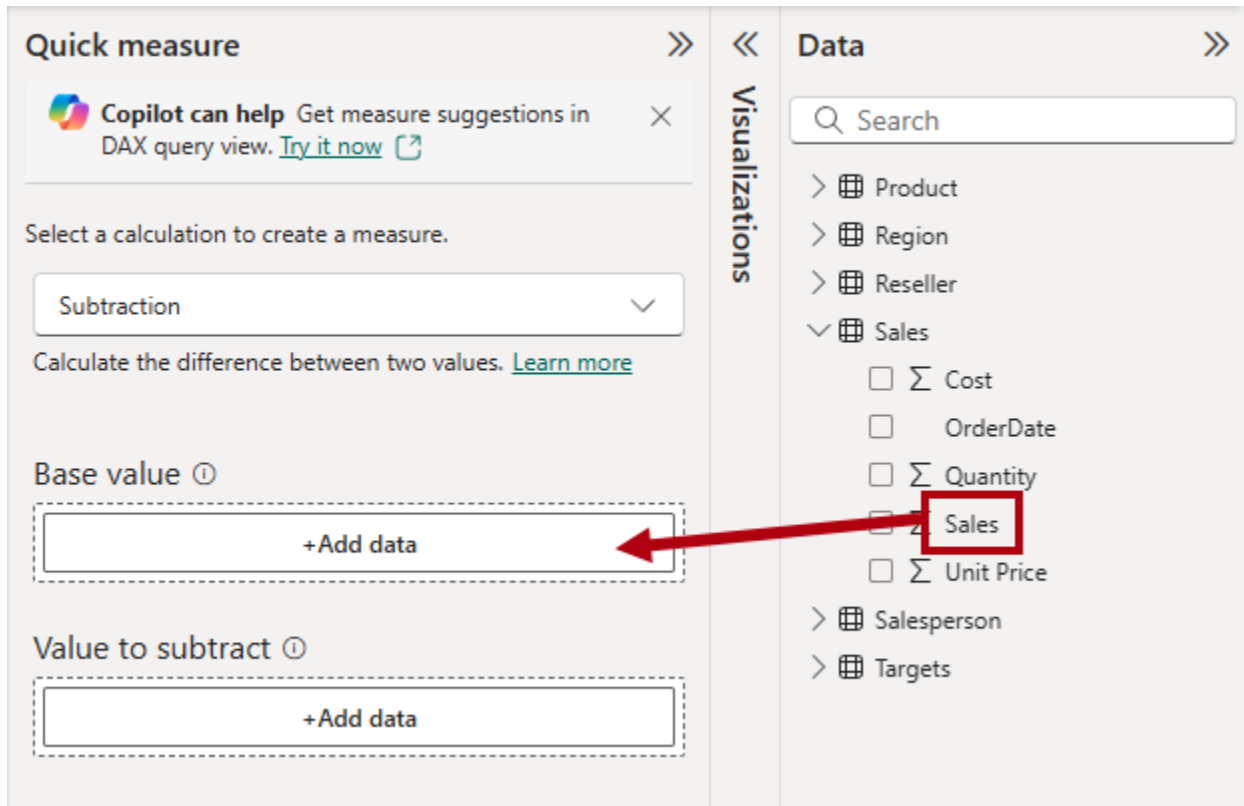
[PL-300-Microsoft-Power-BI-Data-Analyst/Instructions/Labs/02-transform-data-power-bi.md at Main · MicrosoftLearning/PL-300-Microsoft-Power-BI-Data-Analyst](#)



[PL-300-Microsoft-Power-BI-Data-Analyst/Instructions/Labs/03-configure-semantic-model.md at Main · MicrosoftLearning/PL-300-Microsoft-Power-BI-Data-Analyst](#)

- Create model relationships.
- Configure table and column properties.
- Create hierarchies.
- Create quick measures.
- Configure a many-to-many relationship.





Quick measure

 **Copilot can help** Get measure suggestions in DAX query view. [Try it now](#)

Select a calculation to create a measure.

Subtraction

Calculate the difference between two values. [Learn more](#)

Base value ⓘ

+Add data

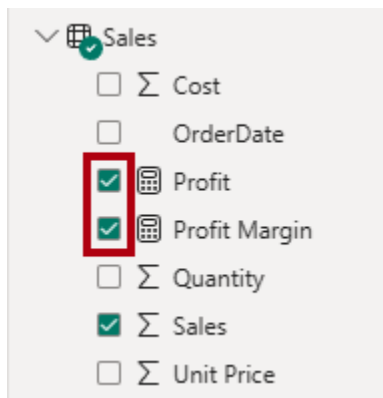
Value to subtract ⓘ

+Add data

Data

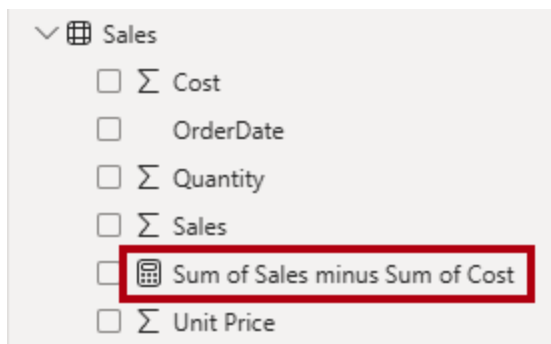
Search

- > Product
- > Region
- > Reseller
- ✓ Sales
 - ☐ Σ Cost
 - ☐ OrderDate
 - ☐ Σ Quantity
 - ☒ Sales
 - ☐ Σ Unit Price
- > Salesperson
- > Targets



✓ Sales

- ☐ Σ Cost
- ☐ OrderDate
- ☒ Profit
- ☒ Profit Margin
- ☐ Σ Quantity
- ☒ Σ Sales
- ☐ Σ Unit Price

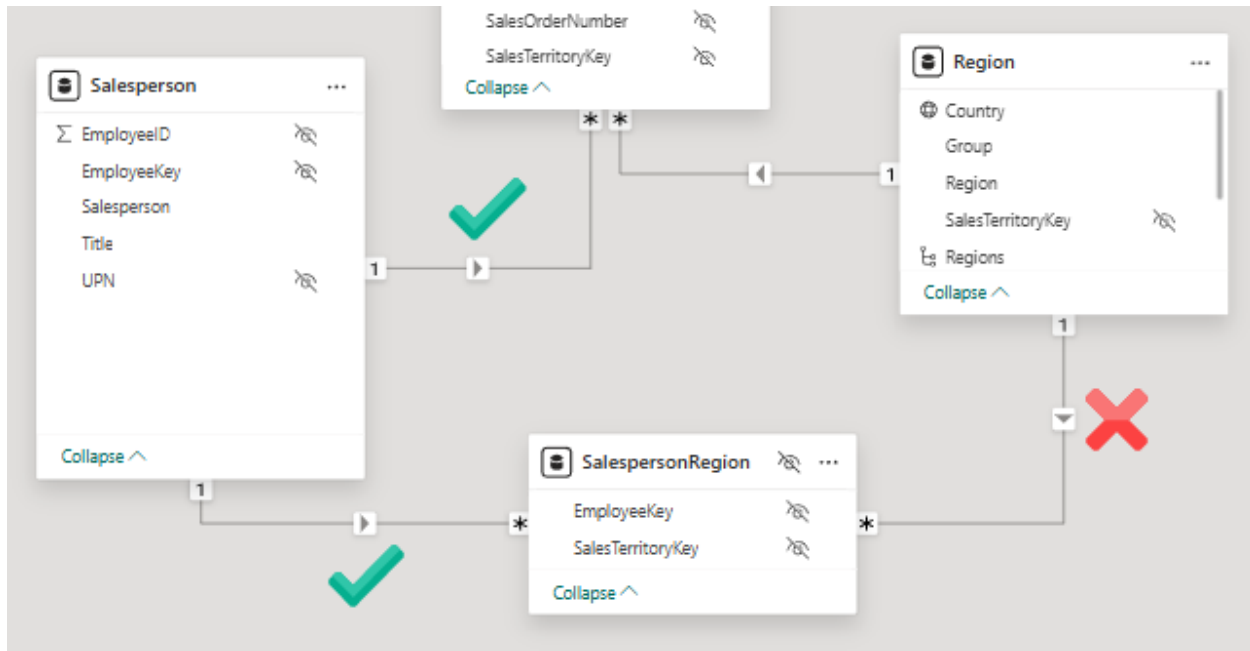


✓ Sales

- ☐ Σ Cost
- ☐ OrderDate
- ☐ Σ Quantity
- ☐ Σ Sales
- ☒ Sum of Sales minus Sum of Cost
- ☐ Σ Unit Price

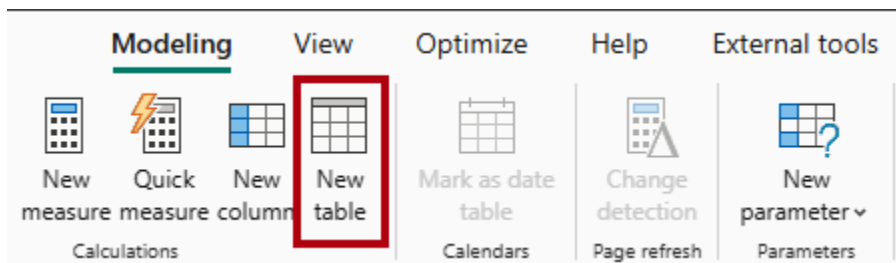
Create a many-to-many relationship between the Salesperson table and the Sales table.

The SalespersonRegion table can be considered to be a bridging table.



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- Create calculated tables.
- Create calculated columns.
- Create measures.



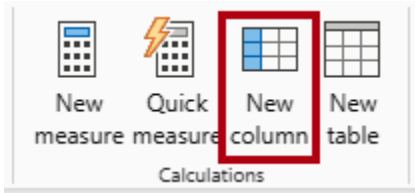
[PL-300-Microsoft-Power-BI-Data-Analyst/Instructions/Labs/04-create-dax-calculations.md at Main · MicrosoftLearning/PL-300-Microsoft-Power-BI-Data-Analyst](#)

Create the Date table

Date =

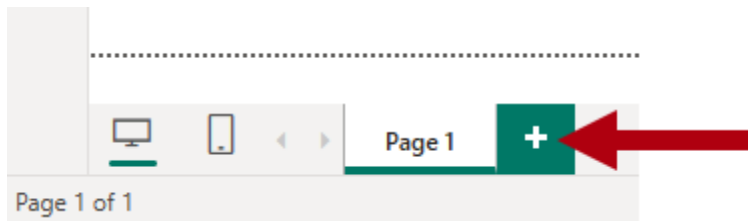
CALENDARAUTO(6)

The CALENDARAUTO function returns a single-column table comprising date values. The "auto" behavior scans all data model date columns to determine the earliest and latest date values stored in the data model. It then creates one row for each date within this range, extending the range in either direction to ensure full years of data is stored.



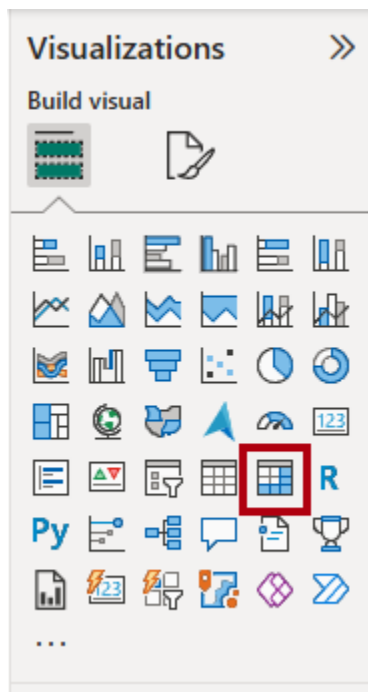
Matrix Visual/Hierarchies

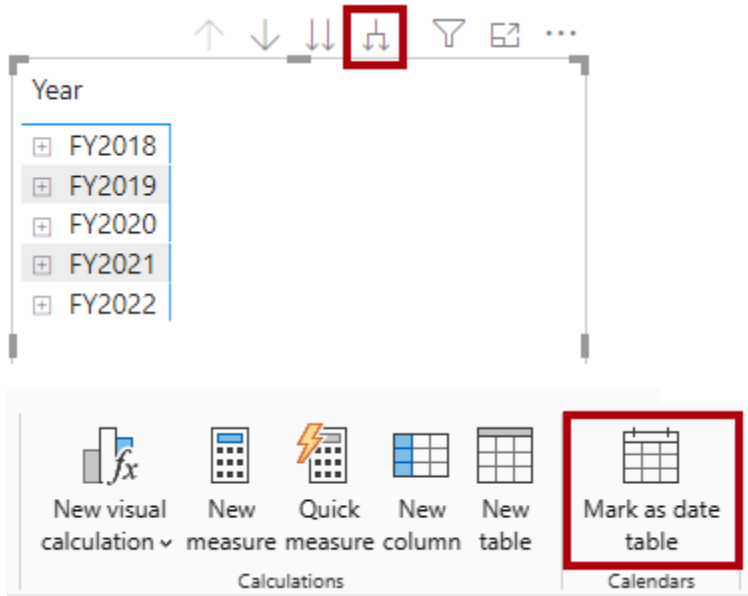
1. To validate the calculations, switch to **Report view**.
2. To create a new report page, select the plus icon next to **Page 1**.



3. To add a matrix visual to the new report page, in the **Visualizations** pane, select the matrix visual type.

Tip: You can hover the cursor over each icon to reveal a tooltip describing the visual type.





Create measures

[PL-300-Microsoft-Power-BI-Data-Analyst/Instructions/Labs/05-modify-dax-filter-context.md](https://www.microsoft.com/learning/PL-300-Microsoft-Power-BI-Data-Analyst/Instructions/Labs/05-modify-dax-filter-context.md) at Main · MicrosoftLearning/PL-300-Microsoft-Power-BI-Data-Analyst

- Use the CALCULATE function to manipulate filter context.

create several measures with DAX expressions that use the CALCULATE function to manipulate filter context.

The CALCULATE function is a powerful function you can use to manipulate the filter context. The first argument takes an expression or a measure (a measure is just a named expression). Subsequent arguments allow modifying the filter context.

Add the Sales All Region measure to the matrix visual.

Group	Country	Region	Sum of Sales	Sales All Region
Europe	France	France	\$4,527,840	\$77,548,570.2
		Total	\$4,527,840	\$77,548,570.2
	Germany	Germany	\$1,877,743	\$77,548,570.2
		Total	\$1,877,743	\$77,548,570.2
	United Kingdom	United Kingdom	\$3,883,043	\$77,548,570.2
		Total	\$3,883,043	\$77,548,570.2
	Total		\$10,288,626	\$77,548,570.2
North America	Canada	Canada	\$13,875,633	\$77,548,570.2
		Total	\$13,875,633	\$77,548,570.2
	United States	Central	\$7,633,387	\$77,548,570.2
		Northeast	\$6,715,354	\$77,548,570.2
		Northwest	\$12,004,822	\$77,548,570.2
		Southeast	\$7,638,607	\$77,548,570.2
		Southwest	\$18,001,116	\$77,548,570.2
		Total	\$51,993,286	\$77,548,570.2
	Total		\$65,868,919	\$77,548,570.2
Pacific	Australia	Australia	\$1,391,025	\$77,548,570.2
		Total	\$1,391,025	\$77,548,570.2
	Total		\$1,391,025	\$77,548,570.2
Total			\$77,548,570	\$77,548,570.2

[PL-300-Microsoft-Power-BI-Data-Analyst/Instructions/Labs/06-use-dax-time-intelligence.md at Main · MicrosoftLearning/PL-300-Microsoft-Power-BI-Data-Analyst](#)

- Use various time intelligence functions to manipulate filter context that specific concerns dates.

Create a sales year-to-date (YTD) measure by using time intelligence functions.

create a sales YoY growth measure by using a variable.

[PL-300-Microsoft-Power-BI-Data-Analyst/Instructions/Labs/07-create-visual-calculations.md at Main · MicrosoftLearning/PL-300-Microsoft-Power-BI-Data-Analyst](#)

- Create and edit visual calculations.
- Use PREVIOUS(), RUNNINGSUM() and MOVINGAVERAGE() functions to create comparison metrics between each fiscal year.
- Use the optional Axis parameter when creating comparison metrics.
- Use the optional Reset parameter to customize cumulative calculations in a multi-levelled axis.